

Proof (sketch): Suppose M_1 is a RM computing χ_{S_1} . From M_1 's program we can construct a RM M_0 to carry out: blue

let $e = R_1$ in $R_1 ::= \ulcorner R_1 ::= 0 ; prog(e) \urcorner ;$

run M_1

Then by assumption on M_1 , M_0 decides membership of S_0 from previous example (i.e. computes χ_{S_0}). **Contradiction.** So no such M_1 exists, i.e. χ_{S_1} is uncomputable, i.e. S_1 is undecidable.